2

Dynamics

The foundational principles of dynamics of bodies were laid down as early as 1666 by Sir Isaac Newton in his historic work "*Principles of Natural Philosophy*". He systematized and summarized the physical phenomena, as we observe it, in three laws of motion.

Newton's First Law

"A body remains in its state of rest or of uniform motion in a straight line unless acted upon by an external force."

From this law we can deduce that if a body is stationary or moving with constant velocity, then the resultant force acting on it will be zero.

The law makes no distinction between objects that are stationary and those moving at a constant velocity as for as the involvement of force is considered.

In both cases, $\sum F = 0$

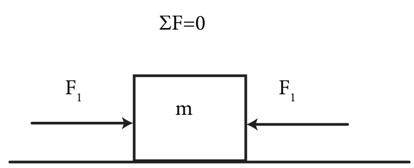


Figure 2.1: Stationary (v=0)

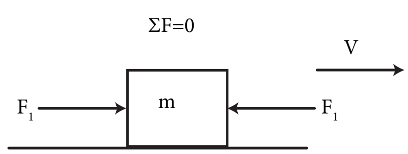


Figure 2.2: Uniform velocity (v=constant)

Newton's Second Law

Rate of change of momentum of a body is proportional to the impressed force and takes place in the direction of the force.

$$F \propto \frac{d}{dt}(mv)$$

where F is the impressed force and mv is mass × velocity (= momentum).

In cases where mass is constant with respect to time,

$$F \propto m\frac{dV}{dt}$$

But

$$\frac{dv}{dt} = a \text{ where } a \text{ is the acceleration.}$$

So, $F \propto ma$ or $F = \lambda ma$

The constant of proportionality λ can be made unity if the force acting on unit mass to produce unit acceleration is taken as the unit of force.

In S.I. units, one Newton is the force required to accelerate one kilogram of mass through one m/s^2.

Thus, when $\lambda = 1$

$$F = ma$$

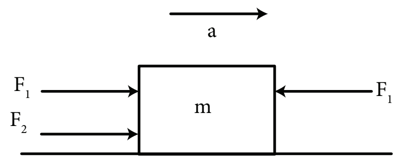


Figure 2.3: Accelerated body
$(\Sigma F \neq 0)\ \Sigma F = (F_1 + F_2) - F_1 = F_2 = ma$

This is the fundamental equation of dynamics.

To produce acceleration in a particular direction, there should be a net force in that direction. In fig. 2.3, acceleration takes place in the direction of the net (resultant) force.

Newton's Third Law

To every action, there is an equal and opposite reaction.

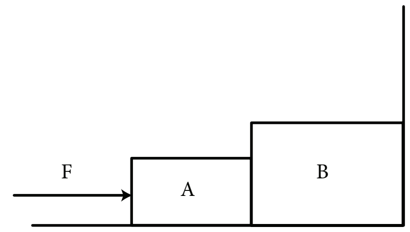


Figure 2.4: Newton's Third Law

Consider the case illustrated in fig. 2.4 In the following case where body A is pushed by a force *F* against body B which in turn, presses against a wall.

The Free Body Diagrams of A and B are shown in fig. 2.5 given below

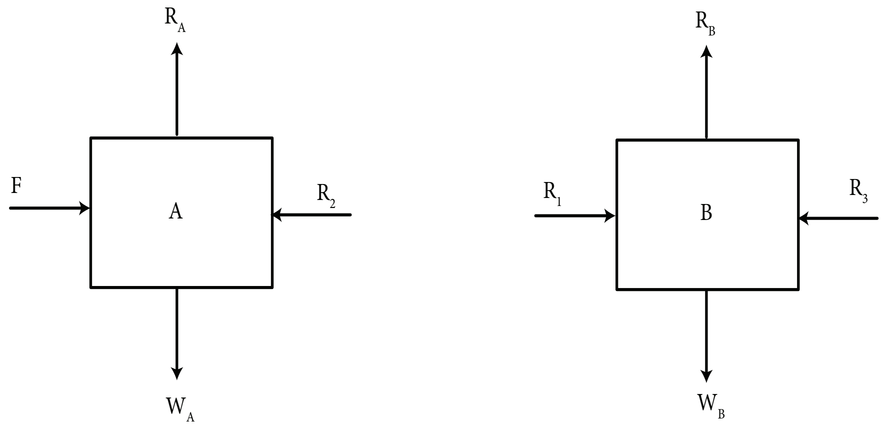


Figure 1.5: (i) Free Body Diagram of A and (ii) Free Body Diagram of B

Here R_1 is the force exerted by A on B.
and R_2 is the force exerted by B on A.
Newton's third law states that $R_1 = R_2$.

Solved Problems

Problem 1

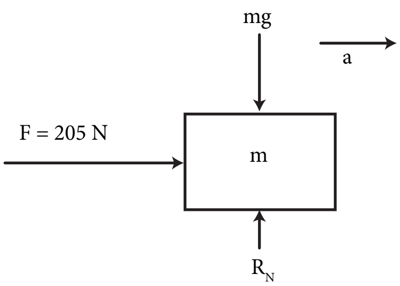
A block of mass 500 kilograms rests on a horizontal plane. A horizontal force of 250 Newtons acts on it for 20 seconds. Find out the velocity acquired by the block at the end of 20 seconds.

Solution

There is a resultant force F in the horizontal direction.

So, there will be an acceleration in the direction of F.

From the equation $F = ma$



$$a = \frac{F}{m} = \frac{250}{500} = 0.5\,m/s^2$$

Velocity acquired by the moving body at the end of 20 seconds,

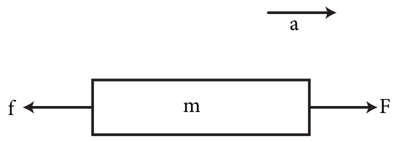
$$\upsilon = u + at$$

$$= 0 + 0.5(20) = 10\,m/s$$

Problem 2

A train of weight 5000 kN acquires a velocity of 36 kmph after 50 seconds. If the track resistance is 4 N per kN of the train's weight, determine the tractive force of the engine.

$$\sum F = F - f$$



Solution

Track resistance, $f = 4\,N \times 5000 = 20000\,N = 20\,kN$

Acceleration of the train,

$$a = \frac{\upsilon - u}{t}$$

where v = final velocity = $36 \times \frac{5}{18} = 10\,m/s$

u = initial velocity = 0

t = time = 50 seconds

$$\therefore, \quad a = \frac{10 - 0}{50} = 0.2\,m/s^2$$

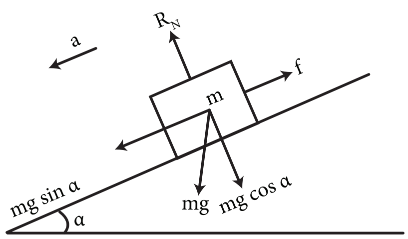
$\sum F = F - f$ where F = tractive force of the engine.

$$\sum F = ma = \frac{5000 \times 10^3}{9.8} \times 0.2 = 102.04\,kN$$

$$\therefore, \; F = \sum F + f = 102.04 + 20 = 122.04\,kN$$

Problem 3

Find out the acceleration of a block of 100 kg sliding down a rough inclined plane of inclination 30° with the horizontal. Given $\mu = 0.25$



Solution

All forces except *mg* are either along or perpendicular to the plane. So it is convenient to resolve *mg* also along and perpendicular to the plane.

There is no acceleration in the perpendicular direction.

So, net force is zero in that direction.

$$R_N = mg\cos\alpha$$

From Newton's equation,

$$\sum F = ma$$

where $\sum F$ is the net force acting downwards.

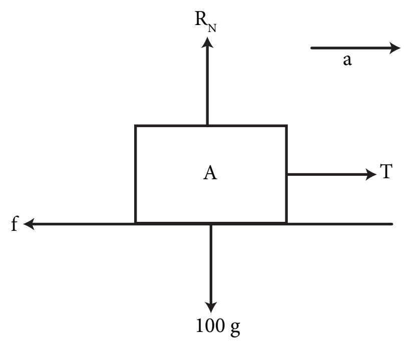
$$50g - T = 50a \qquad \text{(i)}$$

Free Body Diagram of A

There is no acceleration in the vertical direction. So net force in this direction will be zero.

i.e,

$$R_N = 100g \qquad \text{(ii)}$$



Net force towards right produces acceleration a.

$$T - f = ma$$

where $f = \mu R_N = \mu\left(100g\right)$ (from equation ii).

$$\therefore, T - \mu\left(100g\right) = 100a$$

Putting $\mu = 0.25$,

$$T - 25g = 100a \qquad \text{(iii)}$$

Adding equations (i) and (iii),

$$50g - 25g = 150a$$

$$\text{or, } a = \frac{25 \times 9.8}{150} = 1.63 m/s^2$$

From (iii),

$$T = 163 + 25\left(\times 9.8\right) = 408\ N$$

$$\text{ie, } R_N = 100 \times 9.81 \times \cos 30 = 849.6\ N$$

But along the inclined plane, there is acceleration downwards as shown. Resultant force in this direction is

$$\sum F = mg \sin\alpha - f = mg \sin\alpha - \mu R_N$$

$$= \left(100 \times 9.81 \times \sin 30\right) - \left(0.25\right)\left(849.6\right) = 278.1N$$

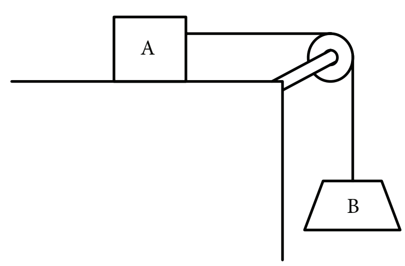
But $\sum F = ma$

So acceleration down the incline is given by

$$a = \frac{\sum F}{m} = \frac{278.1}{100} = 2.781\ m/s^2$$

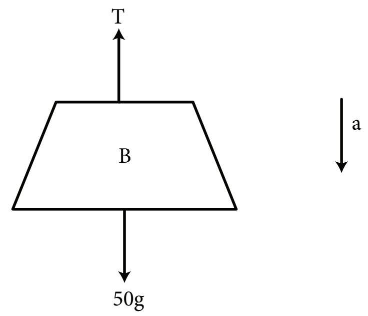
Problem 4

Two blocks A (mass 100 kg) and B (mass 50 kg) are connected by an inextensible string as shown in figure. Find the acceleration of the system and the tension in the string. Given $\mu = 0.25$



Solution

Free Body Diagram of B



From the given figure, It's obvious that the acceleration is downwards.

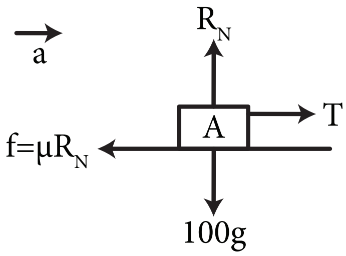
$50g - T = 50a$ (i)

Free Body Diagram of A

No acceleration in the vertical direction. So $\sum F$ in that direction is zero or the forces cancel themselves.

$R_N = 100g$ (ii)

But there is acceleration in the horizontal direction. So,



$$\sum F = m_A a$$

$$T - \mu R_N = m_A a$$

Substitute from (ii),

$T - 0.25(100g) = 100a$ (iii)

Adding (i) and (iii),

$$50g - 0.25(100g) = 150a$$

$$25g = (150)a$$

or

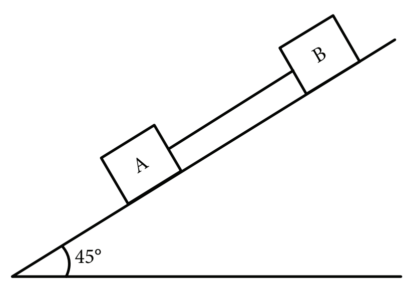
$$a = \frac{150}{25} = 6\, m/s^2$$

From (i),

$$T = 50g - 50a = 50\,(9.8) - 50\,(6) = 190\, N$$

Problem 5

Two blocks A and B of masses 10 kg and 20 kg respectively are connected by a taut string as shown in figure. Calculate the acceleration of the system down the incline and the tension in the string. Given $\mu_A = 0.3$ and $\mu_B = 0.5$



Free Body Diagram of A

Solution

No acceleration in the direction perpendicular to the incline. So,

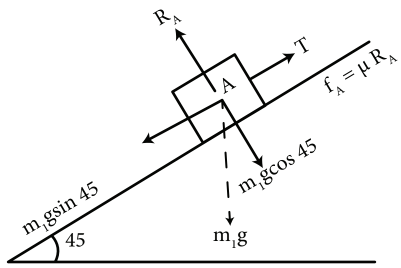
$$R_A = m_1\, g \cos 45 = 10 \times 9.8 \times 0.707 = 69.29\ N$$

Net force along the incline (downward) is given by

$$\sum F = m_1 g \sin 45 - T - f_A = m_1 a$$

But

$$f_A = \mu_A R_A = 0.3 \times 69.29 = 20.79\,N$$



So,

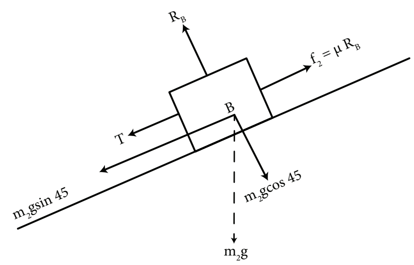
$$10g\,sin\,45 - T - 20.79 = 10\,a \qquad \text{(i)}$$

Free Body Diagram of B

As earlier,

$$R_B = m_2 g\,cos\,45 = 20 \times 9.8 \times 0.707 = 138.57\,N$$

Along the incline, downwards,



$$\sum F = m_2 g\,sin\,45 + T - f_2$$

$$f_2 = \mu_B R_B = 0.5 \times 138.57 = 69.3\,N$$

$$\sum F = m_2 a$$

is,

$$20g\,sin\,45 + T - 69.3 = 20\,a \qquad \text{(ii)}$$

Adding (i) and (ii),

$$30g\,sin\,45 - 90.90 = 30\,a$$

$$a = 3.93\,m/s^2$$

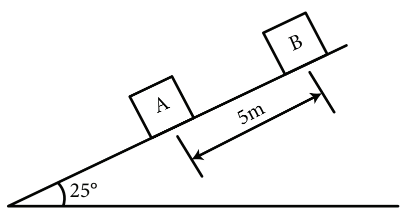
Substituting the value of a in (i), we get

$$T = 9.2\,N$$

Problem 6

Two blocks A and B initially held at rest 5m apart on an inclined plane are released simultaneously. Determine the distance traveled by each block and the time taken before they collide.

Given $\mu_A = 0.25,\ \mu_B = 0.1$



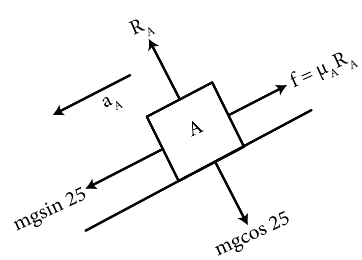
Solution

Let the two blocks be on the verge of collision after time t seconds.

If the distance traveled by A in t seconds is x, then the distance traveled by B in t seconds is $x + 5$.

First we need to find out the accelerations of the two blocks.

Free Body Diagram of A



$$R_A = m_A g\,cos\,25 = m_A \times 9.8 \times 0.906 = 8.88 m_A$$

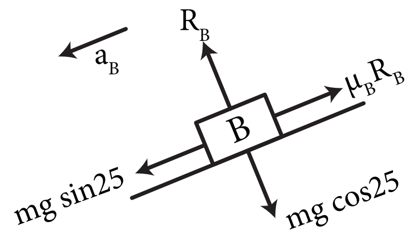
$$m_A g\,sin\,25 - \mu_A R_A = m_A a_A$$

$$m_A\left(9.8 \times 0.423\right) - 0.25\left(8.88\,m_A\right) = m_A a_A$$

Cancel m_A throughout and simplify

$$a_A = 4.15 - 2.22 = 1.93\, m/s^2$$

Free Body Diagram of B



$$R_B = m_B g \cos 25 = 8.88\, m_B$$

$$m_B g \sin 25 - \mu_B R_B = m_B a_B$$

$$m_B (4.15) - (0.1)(8.88)\, m_B = m_B a_B$$

Canceling m_B throughout,

$$a_B = 3.26\, m/s^2$$

Now, distance traveled by A in t seconds

$$x = ut + \frac{1}{2} a_A t^2 = 0 + \frac{1}{2}(1.93)t^2 = 0.97t^2 \qquad (1)$$

Distance traveled by B in t seconds

$$x + 5 = ut\ \frac{1}{2}\, a_B t^2 = 0 + \frac{1}{2}(3.26)t^2 = 1.63t^2 \qquad (2)$$

Equation (2) – (1) gives

$$5 = (1.63 - 0.97)t^2$$

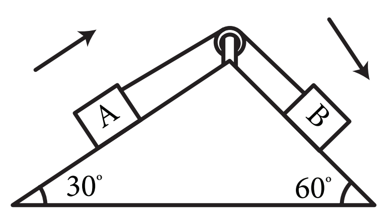
$$\text{or, } t = \sqrt{\frac{5}{0.66}} = 2.75\ s$$

Distance covered by $A = x = 0.97(2.75)^2 = 7.34m$

Distance covered by $B = x + 5 = 12.34m$

Problem 7

Referring to the given figure, find out the acceleration of the connected masses along the incline. Given $m_A = 5kg$, $m_B = 10kg$ and $\mu = 0.25$.



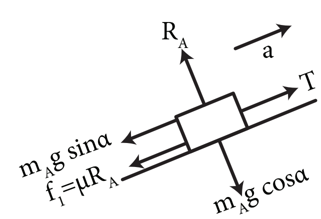
Solution

Free Body Diagram of A

No acceleration in the perpendicular direction.

So,

$$R_A = m_A g \cos\alpha = 5 \times 9.8 \times \cos 30 = 42.4N$$



Along the incline, $\sum F = m_A a$

i.e,

$$T - m_A g \sin\alpha - \mu R_A = m_A a$$

$$T - 5 \times 9.8 \times \sin 30 - 0.25\left(42.4\right) = 5a$$

$$T - 13.9 = 5a \qquad \text{(i)}$$

Free Body Diagram of B

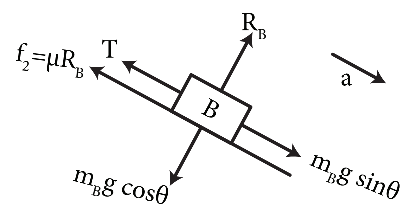
No acceleration in the perpendicular direction. So the forces cancel each other.

$$R_B = m_B g \cos\alpha = 10 \times 9.8 \times \cos 60 = 49\,N$$

$\sum F = m_B a$ along the incline.

i.e,

$$m_B g \sin\theta - T - \mu R_B = m_B a$$



$$10 \times 9.8 \times \sin 60 - T - 0.25\left(49\right) = 10a$$

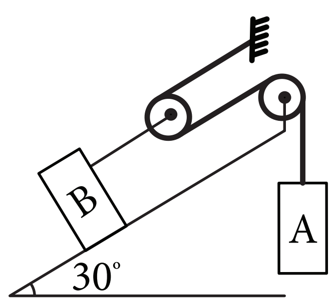
$$72.6 - T = 10a \qquad \text{(ii)}$$

Adding equations (i) and (ii), $58.7 = 15a$

$$a = 3.91 m/s$$

Problem 8

Refer the system shown in adjoining figure. Assuming that the two bodies are released from rest, find out the acceleration of the blocks and tension in the string. Given $m_A = 200$ N and $m_B = 350$ N.



Solution.

From the configuration, it is obvious that the displacement of A will be double the displacement of B for any time interval.

i.e,

$$x_A = 2x_B$$

Differentiating twice,

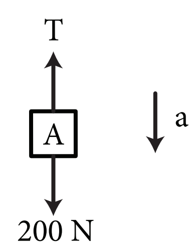
$$a_A = 2a_B \qquad \text{(i)}$$

Equation (i) is the relation between the two accelerations.

As the string is single and continuous, the tension T will be uniform all along (except the part connected to B)

Free Body Diagram of A

$$\sum F = m_A a_A$$



$$200 - T = \left(\frac{200}{9.8}\right) a_A \qquad \text{(ii)}$$

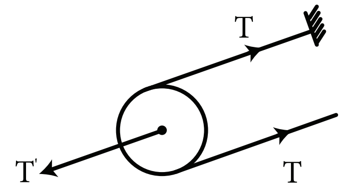
Free Body Diagram of the moving pulley

$$\sum F = ma$$

But the pulley is assumed to be massless.

So $m = o$, or $\sum F = 0$

$T' - 2T = 0$ or $T' = 2T$



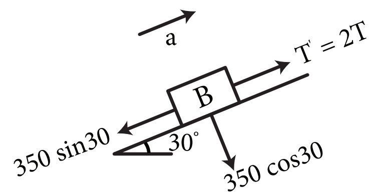
i.e., the tension in the string connected to B is 2T.

Free Body Diagram of B

$$\sum F = m_B a_B$$

$$2T - 350\ sin30 = \left(\frac{350}{9.8}\right) a_B$$

$$2T - 175 = \left(\frac{350}{9.8}\right) a_B \qquad \text{(iii)}$$



Substituting (i) and (ii) and multiplying by 2 gives

$$400 - 2T = \left(\frac{800}{9.8}\right) a_B \qquad \text{(iv)}$$

(iii) + (iv) gives

$$225 = (45.92)\ a_B$$

$$\text{or, } a_B = 4.9 m/s^2$$

$$\text{From (i) } a_A = 4.9 \times 2 = 9.8 m/s^2$$

$$\text{From (ii) } T = 200 - \left(\frac{200}{9.8}\right)(9.8) = ??$$

Problem 9

A balloon of weight *mg* comes down with an acceleration *a*. What amount of weight needs to be thrown out from the balloon to give it the same amount of acceleration upwards?

Solution

Free Body Diagram of the balloon coming down

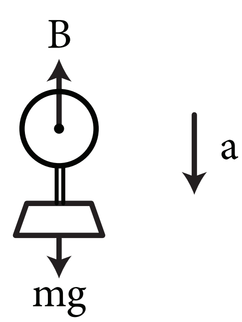
Let *B* be the buoyant force.

$$\sum F = ma$$

$$mg - B = ma \qquad \text{(i)}$$

Free Body Diagram of the balloon going up

Let *m'g* be the weight that is thrown out



$$\sum F = (m - m')a$$

Now,

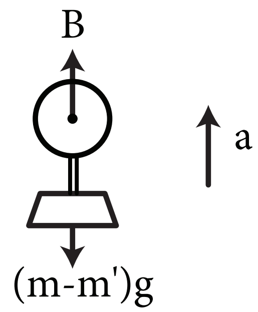
$$B - (m - m')g = (m - m')a$$

or,

$$B - mg + m'g = ma - m'a$$

From (i),

$$B - mg = -ma$$



Substituting this,

$$m'g = 2ma - m'a$$

or,

$$m' = \frac{2ma}{(g+a)}$$

Weight to be thrown out =

$$m'g = \frac{2mga}{(g+a)}$$

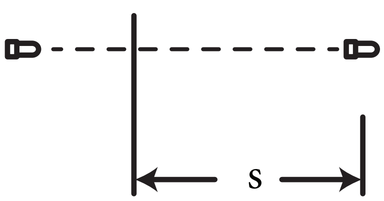
Problem 10

A bullet of mass 30g is fired into the trunk of a huge tree at a velocity of 1000 m/s. Determine how far into the trunk will the bullet penetrate before it stops, if the constant resistance to penetration is 4000 N.

Solution

Let the bullet penetrate through *S* distance into the trunk. If the retardation produced by the resistive force *F* is *a*,

$$F = ma$$



or,

$$a = \frac{F}{m} = \frac{4000}{30 \times 10^{-3}} = 133.33 \times 10^{3}\ m/s^{2}$$

Initial velocity of the bullet when it hit the trunk

$$u = 1000\, m/s$$

Retardation = $133.33 \times 10^{3}\ m/s^{2}$

Final velocity, $\upsilon = 0.$

$$\upsilon^{2} = u^{2} - 2aS$$

$$0 = (1000)^{2} - 2(133.33 \times 10^{3})S$$

or,

$$S = \frac{1000^{2}}{2(133.33 \times 10^{3})} = 3.75\, m$$

Problem 14

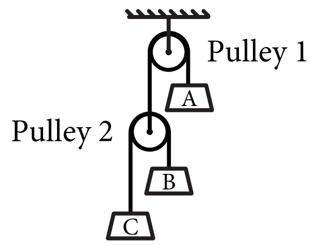
Determine the acceleration of all the three weight assuming weightless and frictionless pulleys.

Given, $W_A = 20\,kg, W_B = 6\,kg$ and $W_C = 8\,kg$

Solution

Let the tensions in the two strings be T_1 and T_2 as shown.
Assume an acceleration 'a', for weight A downwards.

$$20g - T_1 = 20\,a_1 \qquad (1)$$



The moving pulley will have the same acceleration 'a', upwards.
Pulleys are assumed as massless.

So,

$$T_1 - 2T_2 = (0)a$$

or, $T_1 = 2T_2$ (2)

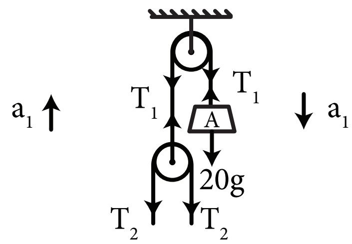
It the acceleration of B be 'a_2' upwards with respect to the pulley 2.

Then 'C' be will have the some acceleration 'a_2' townwards with respect to pulley 2.

But the pulley 2 itself in going up with acceleration a_1.

So the absolute acce acceleration of B is $a_1 + a_2$

and the absolute acceleration of C is $a_1 - a_2$



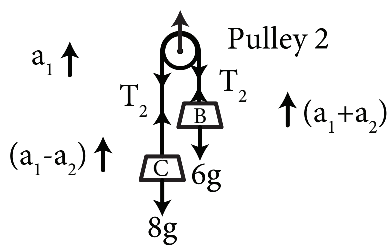
$$T_2 - 8g = 8\,(a_1 - a_2)$$

$$T_2 - 6g = 6(a_1 + a_2)$$

From 1 and 2,

$$20g - 2T_2 = 20a$$

$$T_2 = 10g - 10a$$



Substituting in 3 and 4 we get

$$(10g - 10a_1) - 8g = 8(a_1 - a_2)$$

$$(10g - 10a_1) - 6g = 6(a_1 + a_2)$$

Solving these two simultaneous equations,

we get $a_1 = \dfrac{-5g}{3}$ and $a_2 = 4\,g$

Acceleration of $A = a_1 = \dfrac{-5}{3}g\,m/s^2$

Acceleration of $B = a_1 + a_2 = \dfrac{7g}{3}m/s^2$

Acceleration of $C = a_1 - a_2 = \dfrac{-17}{3}g\,m/s^2$

Motion of a lift

An elevator (or a lift) can accelerate, decelerate and move with constant velocity or can be stationary.

As we have already discussed, the net force is zero ($\sum F = 0$) when the lift is stationary or moving with constant velocity.

When it accelerates or decelerates,the net force will be the inertial force ($\sum F = ma$).

The weight of a man in a lift will always acts downwards and the reaction (R) experienced by the mass from the floor of the lift will be upwards.

The free body diagrams of a man in a lift in the three situations are given.

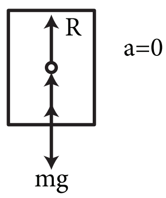


Figure (i) Constant velocity
$mg = R$

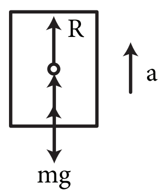


Figure (ii) Accelerating upwards
$R - mg = ma$

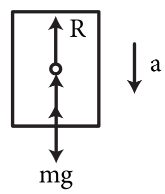


Figure (iii) Accelerating downwards
$mg - R = ma$

In each of the cases referring to the figure, we can write

(i) Constant velocity means $a = 0$.

$$\text{So, } \sum F = 0 \text{ or } mg = R$$

(ii) Accelerating upwards (or decelerating downwards)

$$\sum F = ma \text{ or } R - mg = ma$$

(iii) Accelerating downwards (or decelerating upwards)

$$\sum F = ma \text{ or } mg - R = ma$$

Solved Problems

Problem 12

A lift is going up with a retardation of 2 m/s^2. Find out the pressure exerted on the floor of the lift by a man of weight 700 N.

Solution

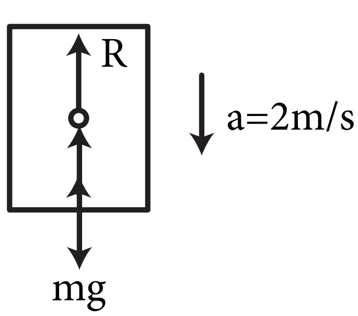
Though the lift is going up, it is decelerating. This means acceleration in the opposite direction.

Free Body Diagram of the man in the lift

$$\sum F = ma$$

$$mg - R = ma$$

$$R = mg - ma = 700 - \left(\frac{700}{9.8}\right)(2) = 557.2\ N$$



The reactions from the floor on the feet of the man is same as the pressure exerted by him on the floor.

So, The pressure exerted = 557.2 N

Problem 13

A man in a lift stands on a weighing machine.The reading in the weighing machine is 800 N when the lift is ascending with an acceleration of 1 m/s^2. What would be its reading

(a) when the lift is going up with a constant velocity?
(b) when it comes down with a retardation of 1 m/s^2 ?
(c) when it comes down with an acceleration of 1 m/s^2 ?

Solution

The reading in the weighing machine will always be equal to the reaction (R) in each situation. Weight of the man *mg* is a constant.

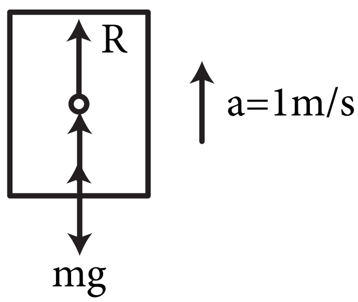
When lift ascends with a = 1 m/s^2

$$\sum F = ma$$

$$R - mg = ma$$

$$800 = m(g + a)$$

$$m = \frac{800}{(9.8 + 1)} = 74.08\,kg$$



Original weight of the man $= mg = (74.08)(9.8) = 726\ N$

(a) When the lift goes up with constant velocity

Constant velocity means $\sum F = 0$

$$R = mg = 726\,N$$

The weighing machine will give the true weight of the man.

(b) When the lift comes down with a retardation of 1 m/s^2

A retardation of 1 m/s^2 is equivalent to an acceleration of 1 m/s^2 upwards. Hence the reading will be the same as given in the question.

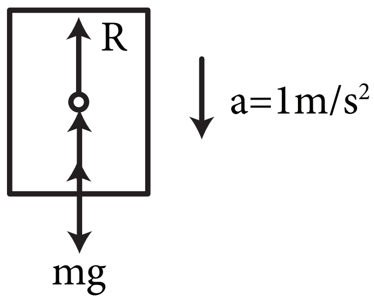
$$R = 800\,N$$

(c) When the lift comes down accelerating at 1 m/s^2

$$\sum F = ma$$

$$mg - R = ma$$

$$R = m\,(g - a) = 74.08\,(9.8 - 1) = 652\,N$$



Problem 14

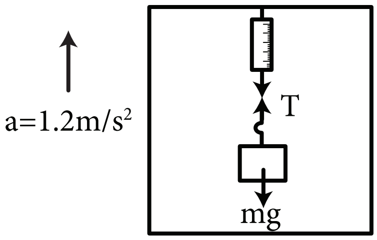
A packet weighing 20 N is hung from a spring balance which is suspended from the roof of a lift. What will be the reading of the spring balance when the lift moves up with an acceleration of 1.2 m/s^2?

Solution

The tension in the string will be the reading of the spring balance in any given situation.

When there is no acceleration (or when stationary) this tension will be equal to the actual weight of the object.

Here the whole system accelerates upwards. So naturally the tension is greater.



$$W = mg = 20\,N$$

$$m = \frac{W}{g} = \frac{20}{9.8}kg$$

$$\sum F = ma$$

$$T - mg = ma$$

$$T = mg + ma = 20 + \left(\frac{20}{9.8}\right)(1.2) = 22.45\,N$$

So the reading will be 22.45 N